Kinematic Drivers of Swing Performance: Sequencing, Plane, and Power

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Research Question

How do kinematic sequencing and swing plane matching relate to each other, and how do these jointly relate to swing performance (bat speed and exit velocity)?

The proper sequencing is proximal to distal, with the angular velocity of the pelvis peaking first followed by the torso then lead arm and hand (Don, 2026). Each segment should reach a peak of higher magnitude than the one before it, acting like a whip. We look at the time between each segment's peak (lag) to determine if a batter's swing follows the sequence.

MLB pitches typically descend towards home plate at a vertical angle of -3° to -7° (Crizer, 2023), while the ideal vertical attack angle is between 5° and 20° to get the most offensive value (MLB, 2025). This gives an ideal vertical plane delta of around 10° to 30°.

Key Inferences

The results support kinematic sequencing as the foundation of a powerful swing. A clean proximal to distal sequence (Batter 1) sees high collision efficiency, with high bat speed producing a high exit speed even by MLB standards. When the sequence becomes dominated by the upper half, raw bat speed isn't as reliable.

Batter 2 demonstrates that high bat speed can be generated primarily by upper body rotational force, but without the lower body stability provided by proper sequencing originating at the pelvis, barrel control may suffer, leading to whiffs. The angular velocity plots highlight these mechanical contrasts, showing Batter 1's distinct, staggered curves peaking in order, compared to Batter 2's overlapping curves with the torso going early. Based on the differences between Batter 1 and Batter 2's swing (both have good planes, but different sequencing), we can hypothesize that sequencing may be relevant here.

The contrast between Batter 4 and Batter 5 highlights swing plane matching as a potential compensatory mechanism for poor sequencing. Even with flawed sequencing, Batter 4 flattened their swing to the approaching pitch and achieved contact with a strong exit speed. When a batter breaks the sequence *and* doesn't match the pitch's plane well (like Batter 5), both power generation and contact quality suffered.

Although mentioned in the research question, there is no evidence that kinematic sequencing enables better plane matching. The results do suggest that while a flawed kinematic sequence doesn't inherently prevent a batter from establishing a good swing plane, plane matching is a crucial compensatory mechanism in the absence of proper sequencing in order to achieve solid contact. After further investigation (see Recommended Next Steps), coaches can leverage this insight to design targeted interventions, focusing on sequencing drills for power-limited hitters, while prioritizing barrel-path precision for those relying on compensatory plane matching.

Uncertainty & Limitations

This analysis reveals clear contrast between these batters, but the relevance of these findings is limited to this specific case study and can't be generalized due to the small sample size of four analyzed swings. There is no statistical power behind these findings.

While pitch tracking was available, the absence of a 3D strike-zone prevents the mapping of pitch location relative to each batter's zone. So we cannot determine whether extreme vertical attack angles or large plane deltas are due to a mechanical breakdown in the swing, adaptation to an extreme pitch, or other factors. In addition to this, the motion-capture data only accounts for physical output rather than muscle activation, which limits our ability to look at force generation opposed to natural movement flow.

Applications

- Player evaluation: evaluating batters strictly by peak bat speed can hide mechanical breakdowns, so performance staff should decouple power generation (sequence efficiency) from delivery (swing plane matching) to differentiate power-limited from path-limited (generating high speed but missing the pitch plane)
- Training intervention: coaches can tailor drills to a batter's profile. Power-limited batters can work on force transfer while path-limited batters can work on pitch location adjustment and precision
- Monitoring: looking at the exact timing of pelvis-torso and torso-hand lags throughout the season can help detect a change in mechanics that might be due to something like fatigue or injury

Recommended Next Steps

- Data collection: collect more swings beyond the four batters and multiple trials per batter in a controlled environment to allow for more powerful statistical testing and modelling
- Once a baseline is established (more data collected):
    - Develop player profiles, identify any potential mechanical breakdowns
    - Examine whether bat speed is more sensitive to breakdowns in the pelvis-torso segment or torso-hand segment
    - Identify which part of the sequence is the first to lapse under fatigue (workload management)

Resources

AI Usage: Generative AI (Gemini) was used as an editing and debugging tool, specifically for refining research question wording, generating project title options, condensing text, and troubleshooting R code.

Crizer, Z. (2023, April 19). *How vertical approach angle, or vaa, is helping MLB pitchers understand and improve the shape of their fastballs*. Yahoo! Sports. https://sports.yahoo.com/how-vertical-approach-angle-or-vaa-is-helping-mlb-pitchers-understand-and-improve-the-shape-of-their-fastballs-224115419.html

Don, A. (2026, June 3). *What is on plane efficiency in baseball?*. Blast Motion. https://blastmotion.com/blog/what-is-on-plane-efficiency-in-baseball/

Ideal attack angle: Glossary. MLB.com. (n.d.). https://www.mlb.com/glossary/statcast/ideal-attack-angle